IGCSE 0607 Practice Questions — E2.7 — Combined Questions & Markscheme

Sequences

Total Questions: 18 — **Total Marks:** 111 Marks

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1 Here are the first four terms of a sequence.

5 12 19 26

(a) Write down the next term in the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

2 Here are the first four terms of a sequence.

5 11 17 23

(a) Find the next term in this sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

3 **(a)** Find the nth term of the sequence.

5 9 13 17 21

..... **[2]**

(b) Find the nth term of the sequence.

2 6 18 54 162

..... **[2]**

4 The first four terms of a sequence are

7 11 15 19

(a) Write down the next two terms of the sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

nth term = ... **[2]**

(c) Explain why 102 is not a term in this sequence.

..... **[2]**

5 **(a)** Sequence A is a linear sequence.

$$5 \quad 9 \quad 13 \quad 17 \quad \ldots$$

(i) Find the next two terms of sequence A.

..... **[1]**

(ii) Find the nth term of sequence A.

..... **[2]**

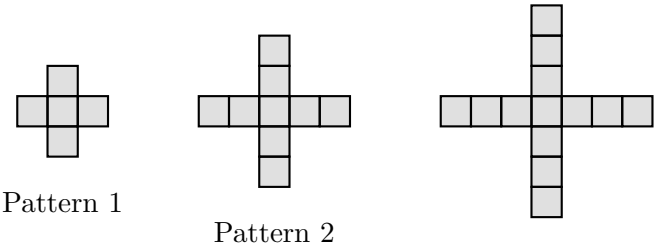
(b) Sequence B is a geometric sequence.

$$2 \quad 6 \quad 18 \quad 54 \quad \ldots$$

Find the nth term of sequence B.

..... **[2]**

6 The diagram shows a sequence of patterns made from small squares.



Pattern 1

Pattern 2

Pattern 3

(a) Find the number of squares in Pattern 4.

..... **[1]**

(b) Find an expression, in terms of n, for the number of squares in Pattern n.

..... **[2]**

(c) Which pattern has exactly 141 squares?

Pattern ..... **[2]**

7 The first four terms of a sequence are

$$-1 \quad 4 \quad 13 \quad 26$$

(a) Find an expression for the nth term of the sequence.

..... **[3]**

(b) Find the value of n when the nth term is 433.

$n =$ ..... **[3]**

8 The first five terms of a sequence, T_n, are given below.

$$4 \qquad 14 \qquad 34 \qquad 68 \qquad 120$$

(a) By finding the first, second and third differences, show that the nth term of the sequence is a cubic expression in n.

[2]

(b) Find an expression for T_n, the nth term of the sequence.

$T_n =$ ..... **[5]**

9 Here are the first five terms of a sequence.

4 11 22 37 56

(a) Work out the next term of the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[3]**

10 **(a)** The nth term of a sequence is $n^2 - 3n + 5$.
Work out the first three terms of this sequence.

1st term = ... , 2nd term = ... , 3rd term = ...
[2]

(b) The first four terms of another sequence are

2 9 20 35

Find an expression for the nth term of this sequence.

..... **[3]**

11 The first four terms of a sequence, T, are

$$2 \quad 9 \quad 22 \quad 41$$

(a) Find an expression for the nth term, T_n.

$T_n =$ ..... **[3]**

(b) A second sequence is given by the formula $U_n = 2^n + 3$.
Work out the value of U_5.

$U_5 =$ ..... **[1]**

(c) Find the value of n for which $U_n = 131$.

$n =$ ..... **[2]**

12 The first four terms of a sequence are

$$-1 \quad 2 \quad 9 \quad 20$$

(a) Find the nth term of this sequence.

..... **[3]**

(b) The nth term of another sequence is given by $3^n - 2$.
Find the value of the term in this sequence that is closest to 1000.

..... **[3]**

13 The first five terms of a sequence are

2 9 20 35 54 ...

(a) Find the next term.

..... **[1]**

(b) Find the nth term of the sequence.

..... **[3]**

(c) The kth term of the sequence is 464.
Find the value of k.

$k =$ ..... **[3]**

14 Consider the following three sequences, P, Q and R.
Sequence P:

$$2 \quad 9 \quad 28 \quad 65 \quad 126 \quad \ldots$$

Sequence Q:

$$3 \quad 8 \quad 15 \quad 24 \quad 35 \quad \ldots$$

Sequence R:

$$\frac{2}{3} \quad \frac{9}{8} \quad \frac{28}{15} \quad \frac{65}{24} \quad \frac{126}{35} \quad \ldots$$

(a) Find the nth term of sequence P.

..... **[2]**

(b) Find the nth term of sequence Q.

..... **[2]**

(c) Write down the nth term of sequence R.

..... **[1]**

(d) Find the 12th term of sequence R, giving your answer as a fraction in its simplest form.

..... **[2]**

15 **(a)** The first five terms of a sequence are

$$-1 \quad 8 \quad 33 \quad 80 \quad 155$$

Use the method of differences to find an expression for the nth term of this sequence.

..... **[4]**

(b) An exponential sequence has nth term $P_n = 5 \times 3^{n-1}$.
The kth term of this sequence is 3645.
Work out the value of k.

$k =$ ..... **[2]**

(c) The nth term of sequence T is $T_n = \dfrac{2n-1}{3n+2}$.
Write down the value that T_n approaches as n becomes very large.

..... **[1]**

16 **(a)** A sequence has the following first four terms:

$$1 \quad 15 \quad 53 \quad 127$$

Use the method of differences to find an expression for the nth term.

nth term = ... **[4]**

(b) Another sequence, V, has its nth term defined by

$$V_n = \frac{n^2 + 1}{2^n}$$

(i) Find the values of V_1, V_2 and V_3.
Give each fraction in its simplest form.

V_1 = ... , V_2 = ... , V_3 = ... **[2]**

(ii) Work out the exact value of $V_5 - V_4$.

..... **[3]**

17 **(a)** The first five terms of a sequence are

4 5 14 37 80

Use the method of differences to find an expression for the nth term.

..... **[4]**

(b) The nth term of a different sequence is given by $T_n = a \times 2^n + b$.
The first term of this sequence is 10 and the third term is 34.

Find the values of a and b.

$a =$ ..... , $b =$ ..... **[4]**

(c) Calculate the 12th term of the sequence described in part (b).

..... **[2]**

18 The first five terms of three sequences, A, B and C, are shown below.

Sequence A: 5 9 13 17 21

Sequence B: 2 6 20 50 102

Sequence C: $\frac{5}{2}$ $\frac{3}{2}$ $\frac{13}{20}$ $\frac{17}{50}$ $\frac{7}{34}$

(a) Find the nth term of sequence A.

..... **[2]**

(b) **(i)** Work out the 6th term of sequence B.

..... **[1]**

(ii) Find an expression for the nth term of sequence B.

..... **[4]**

(c) **(i)** Work out the 6th term of sequence C, giving your answer as a fraction in its simplest form.

..... **[1]**

(ii) Find an expression for the nth term of sequence C.

..... **[2]**

1 **(a)** 33 **[B1]**

(b) $7n + k$ for any integer k **[M1]**

$7n - 2$ **[A1]**

(c) $7n - 2 = 150$

$7\text{n} = 152 whichdoesnotgiveanintegervaluefor\text{n}.[B1]$

[Back to Question] *[Total Marks: 4]*

2 **(a)** 29 **[B1]**

(b) $6n + c$ or finding common difference of 6 **[M1]**

$6n - 1$ **[A1]**

(c) Sets $6n - 1 = 150 \implies 6n = 151$ and states that 151 is not divisible by 6 (or n is not an integer) **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Common difference is 4 **[M1]**

$4n + 1$ **[A1]**

(b) Identifies common ratio is 3 **[M1]**

$2 \times 3^{n-1}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** $23, 27$ **[B1]**

(b) $4n + k$ (for any k) **[M1]**

$4n + 3$ **[A1]**

(c) Equates $4n + 3 = 102$ and attempts to solve for n **[M1]**

$n = 24.75$, which is not a whole number/integer **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** **(i)** 21 and 25 **[B1]**

(ii) Identifies common difference of 4 leading to $4n$ **[M1]**

$4n + 1$ **[A1]**

(b) Identifies common ratio of 3 **[M1]**

$2 \times 3^{n-1}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** 17 **[B1]**

(b) $4n + c$ where c is a constant **[M1]**

$4n + 1$ **[A1]**

(c) $4n + 1 = 141$ or $141 - 1 = 140$ **[M1]**

35 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** Identifying the second differences are constant 4 **[M1]**

Deducing the n^2 coefficient is 2, or setting up $2n^2 + bn + c$ **[M1]**

$2n^2 - n - 2$ **[A1]**

(b) $2n^2 - n - 2 = 433$ **[M1]**

$2n^2 - n - 435 = 0$ leading to correct factorisation $(2n + 29)(n - 15) = 0$ or correct substitution into the quadratic formula **[M1]**

15 (must reject negative solution if given) **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** First differences: $10, 20, 34, 52$ **[M1]**

Second differences: $10, 14, 18$
Third differences: $4, 4$
Since the third difference is a non-zero constant, the sequence is cubic. **[A1]**

(b) Uses $6a = 4 \implies a = \frac{2}{3}$ **[M1]**

Uses $12a + 2b = 10 \implies 12(\frac{2}{3}) + 2b = 10 \implies 8 + 2b = 10 \implies b = 1$ **[M1]**

Uses $7a + 3b + c = 10 \implies 7(\frac{2}{3}) + 3(1) + c = 10 \implies c = 10 - 3 - \frac{14}{3} = \frac{7}{3}$ **[M1]**

Uses $a + b + c + d = 4 \implies \frac{2}{3} + 1 + \frac{7}{3} + d = 4 \implies 4 + d = 4 \implies d = 0$ **[M1]**

$\frac{2}{3}n^3 + n^2 + \frac{7}{3}n$ **[A1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** 79 **[B1]**

(b) Calculates second differences to be 4, or equates $2a = 4$ **[M1]**

Deduces that the sequence contains $2n^2$ **[M1]**

$2n^2 + n + 1$ **[A1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** One correct term evaluated **[M1]**

$3, 3, 5$ **[A1]**

(b) Second differences are 4 **[M1]**

Identifies $2n^2$ as part of the expression **[M1]**

$2n^2 + n - 1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

11 (a) Finds first differences of $7, 13, 19$ and constant second differences of 6 **[M1]**

Deduces the quadratic term is $3n^2$ **[M1]**

$3n^2 - 2n + 1$ **[A1]**

(b) 35 **[B1]**

(c) $2^n + 3 = 131$ leading to $2^n = 128$ **[M1]**

7 **[A1]**

[Back to Question] *[Total Marks: 6]*

12 (a) Finds first differences: $3, 7, 11$ and second differences: $4, 4$ **[M1]**

Deduces the term involves $2n^2$ **[M1]**

$2n^2 - 3$ **[A1]**

(b) $3^n - 2 \approx 1000 \implies 3^n \approx 1002$ **[M1]**

Uses logarithms or trial and improvement: $n = 6 \implies 3^6 - 2 = 729 - 2 = 727$; $n = 7 \implies 3^7 - 2 = 2187 - 2 = 2185$ **[M1]**

727 **[A1]**

[Back to Question] *[Total Marks: 6]*

13 (a) 77 **[B1]**

(b) Finds first differences $(7, 11, 15, 19)$ and constant second difference of 4 **[M1]**

Deduces the coefficient of n^2 is 2 **[A1]**

Subtracts $2n^2$ from the sequence terms to identify the linear sequence $n - 1$ **[M1]**

$2n^2 + n - 1$ **[A1]**

(c) Sets up the equation $2k^2 + k - 1 = 464$ **[M1]**

Rearranges to form a quadratic equal to zero: $2k^2 + k - 465 = 0$ **[M1]**

15 **[A1]**

[Back to Question] *[Total Marks: 7]*

14 (a) Recognises the relationship to cube numbers $(1, 8, 27, 64, 125)$ **[M1]**

$n^3 + 1$ **[A1]**

(b) Identifies second difference is 2, or relates to square numbers **[M1]**

$n^2 + 2n$ or $(n+1)^2 - 1$ **[A1]**

(c) $\dfrac{n^3+1}{n^2+2n}$ **[B1]**

(d) Substitutes $n = 12$ into their expression for sequence R, yielding $\dfrac{1728+1}{144+24}$ or $\dfrac{1729}{168}$ **[M1]**

Simplifies the fraction correctly to $\dfrac{247}{24}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 (a) Finds third differences are 6 (first diffs: $9, 25, 47, 75$; second diffs: $16, 22, 28$; third diffs: 6) **[M1]**

Identifies n^3 as the cubic term **[B1]**

Subtracts n^3 from the sequence to get a quadratic sequence $(-2, 0, 6, 16, 30)$ and finds its second differences are 4 to give $2n^2$ **[M1]**

$n^3 + 2n^2 - 4n$ **[A1]**

(b) $5 \times 3^{k-1} = 3645$

$3^{k-1} = 729$ [M1]

7 **[A1]**

(c) $\dfrac{2}{3}$ or 0.667 **[B1]**

[Back to Question] *[Total Marks: 7]*

16 (a) Finds first differences $14, 38, 74$ and second differences $24, 36$ **[M1]**

Finds third difference of 12 leading to a cubic term of $2n^3$ **[M1]**

Evaluates the sequence of residuals $T_n - 2n^3$ to find a constant difference of -1 **[M1]**

$2n^3 - 1$ **[A1]**

(b) (i) $V_1 = 1$, $V_2 = \dfrac{5}{4}$, $V_3 = \dfrac{5}{4}$ **[B2]**

Award B1 if any two are correct **[B1]**

(ii) Substitutes $n = 5$ to find $V_5 = \dfrac{25+1}{32} = \dfrac{13}{16}$ **[B1]**

Substitutes $n = 4$ to find $V_4 = \dfrac{16+1}{16} = \dfrac{17}{16}$ **[B1]**

Evaluates $\dfrac{13}{16} - \dfrac{17}{16} = -\dfrac{1}{4}$ (accept -0.25) **[A1]**

[Back to Question] *[Total Marks: 9]*

17 (a) Correct first differences $1, 9, 23, 43$ and second differences $8, 14, 20$ and third differences 6 **[M1]**

Deducing the cubic term coefficient is $a = 6 \div 6 = 1$ **[B1]**

Setting up equations to solve for other coefficients, e.g. $12a + 2b = 8$ **[M1]**

$n^3 - 2n^2 + 5$ **[A1]**

(b) $2a + b = 10$ **[B1]**

$8a + b = 34$ **[B1]**

Correct method to solve simultaneous equations, e.g. subtracting to get $6a = 24$ **[M1]**

$a = 4, b = 2$ **[A1]**

(c) Substituting $n = 12$ into their $4 \times 2^n + 2$ **[M1]**

16386 **[A1]**

[Back to Question] *[Total Marks: 10]*

18 **(a)** $4n + 1$ **[B2]**

If $4n + k$ seen, award **[M1]**

(b) **(i)** 182 **[B1]**

(ii) Identifies third differences are constant (6), so term contains n^3 **[M1]**

Subtracts n^3 terms $(1, 8, 27, 64, 125)$ from the sequence **[M1]**

Remainder sequence is $1, -2, -7, -14, -23$ **[M1]**

Deduces the remainder is $-n^2 + 2$ **[M1]**

$n^3 - n^2 + 2$ **[A1]**

(c) **(i)** $\frac{25}{182}$ **[B1]**

(ii) Recognises the pattern is Sequence A divided by Sequence B **[M1]**

$\frac{4n + 1}{n^3 - n^2 + 2}$ **[A1]**

[Back to Question] *[Total Marks: 10]*